# Tutorial - 7

In this tutorial, you will work on the following probelms: (1) derivative calculation using MPI for several explicit finite differnce numerical schemes (2) understand and demonstrate a few advanced collective communication calls (3) install NVidia HPC SDK so as to enable compilation of OpenACC programs using *pgcc* or *pgfortran*.

1. Consider the following

$$u(x) = x^3 - \sin(5x), \tag{1}$$

and $x = [0, 3]$. Write an MPI program to compute the first derivative of $u(x)$ that is $du/dx$ over the range of $x$ given using first, second and fourth-order accurate formulae as given below. You can use first-order accurate formulae on the boundary and near-boundary points when using higher-order schemes for the interior points. Use grid sizes of $\Delta x = 0.01$ and $0.001$. Test your code using 2, 4 and 8 processes. Compare the results you obtain using the analytical solution and make sure the numerical solution obtained using your MPI code is correct. Plot the derivative function on a single graph for both the analytical solution and the numerical solution (for one of the schemes). You can make use of virtual points (ghost points) to store the data belonging to neighboring processes that are required while calculating the derivatives at the processor boundary points and near boundary points. Convince yourself that the error (between the analytical and the numerical) is much smaller with higher-order methods. The finite difference formulae are given below:

$$f'(x) = \frac{f(x_{i+1}) - f(x_i)}{\Delta x} + O(\Delta x) \tag{2}$$

$$f'(x) = \frac{f(x_i) - f(x_{i-1})}{\Delta x} + O(\Delta x) \tag{3}$$

$$f'(x) = \frac{f(x_{i+1}) - f(x_{i-1})}{2\Delta x} + O(\Delta x^2) \tag{4}$$

$$f'(x) = \frac{f(x_{i-2}) - 8f(x_{i-1}) + 8f(x_{i+1}) - f(x_{i+2})}{12\Delta x} + O(\Delta x^4). \tag{5}$$

$$\tag{6}$$

2. Develop MPI programs to demonstrate collective functions such as MPI_Gatherv, MPI_Scatterv, MPI_Alltoall, and the timing functions MPI_Wtime, MPI_Wtick. Convince yourself that these programs are producing the intended output by running them.

3. Install NVidia HPC SDK and use pgcc for compilation of programs on GPU. Try compiling and running the Hello World program that we have seen in the class.